# AIR QUALITY MONITORING SYSTEM

23BEC1378 – LAKSHMI SANJEEV

23BLC1060 – NEHA RIONA ALICE K

23BLC1224 – MIRUNALINI A

# PROJECT OVERVIEW

## WHY?

- Air quality is the degree to which the air is safe for humans and the environment. Good air quality means the air is free of harmful substances, and clear with only small amounts of solid particle and chemical pollutants. Poor air quality is often hazy and dangerous to health and the environment. Knowing the air quality can help you plan for safe and healthy physical activity.
- Breathing problems and lung cancer are frequently recorded in places with high toxicity.
- Getting an understanding of the air quality can help people maintain good health by taking appropriate measures.

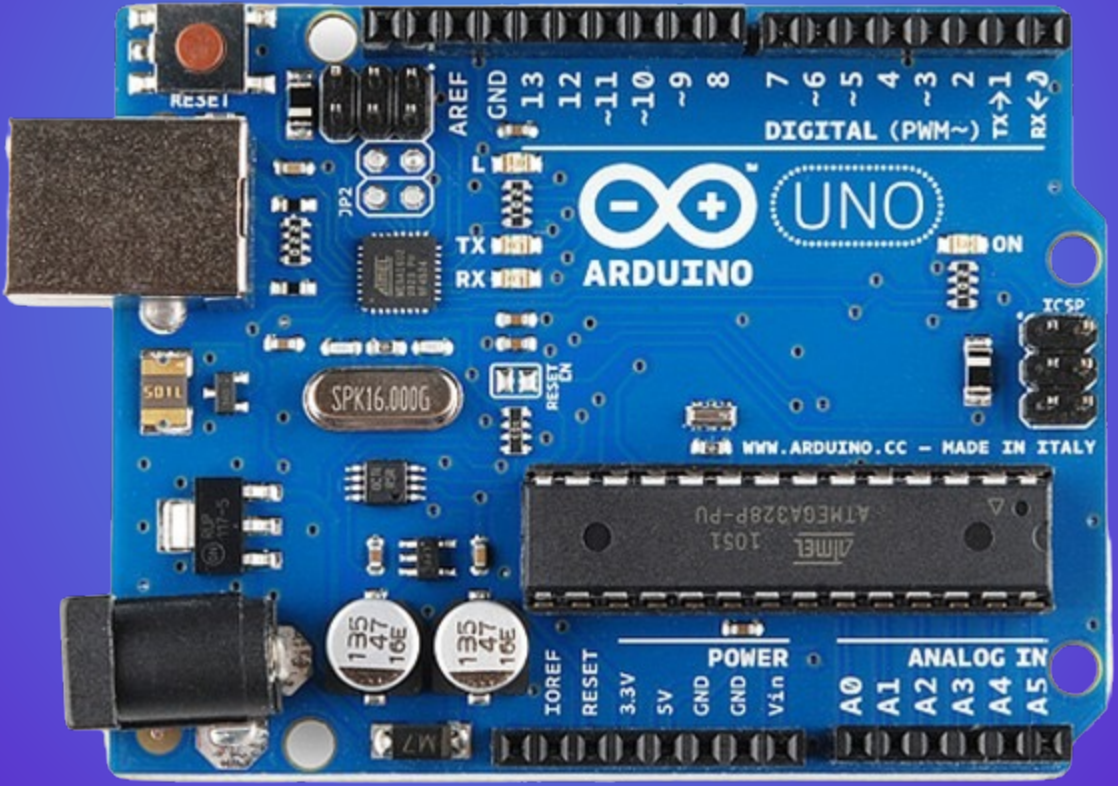
# PROJECT OVERVIEW

## WHAT?

- The main objective of the project is using SB-microcontrollers and various sensors to detect various types of gases, particles and smoke in the air to determine the quality of air around us, thus helping us to plan our surroundings accordingly.
- By the end of this project, with the resources in hand, we will be able to determine and display if the air around us is safe enough or not.

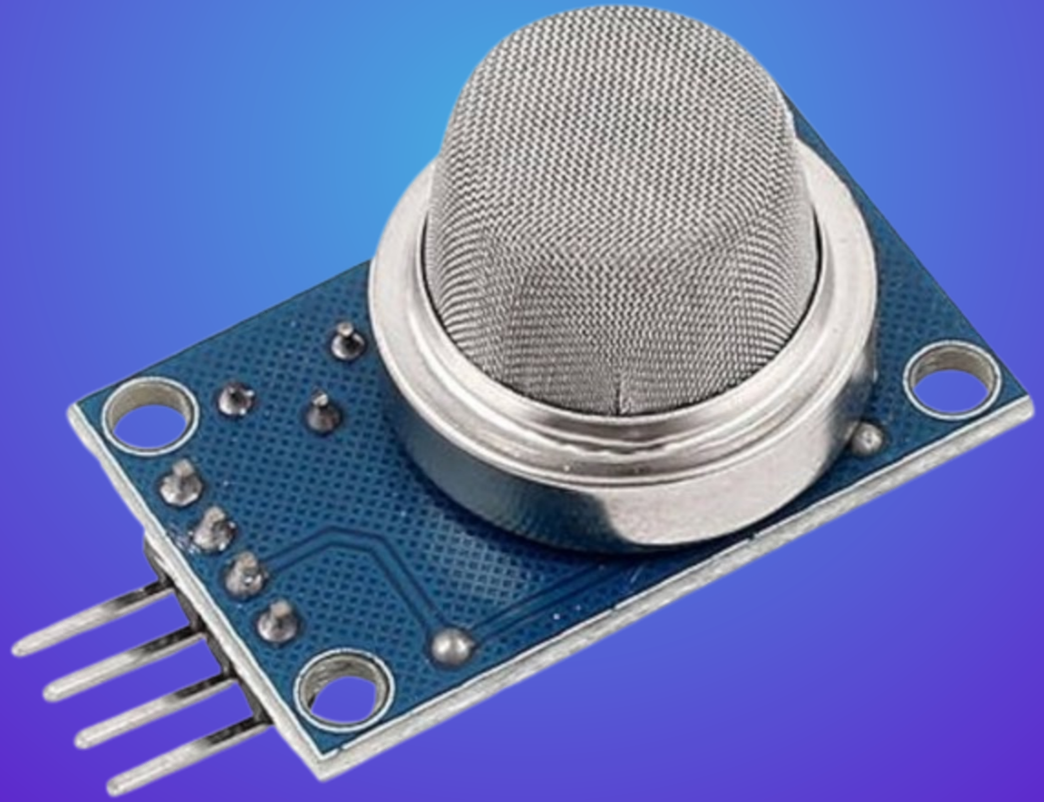
# COMPONENTS

# Single-Board Microcontrollers & Software IDE



# Gas Sensors – MQ Series

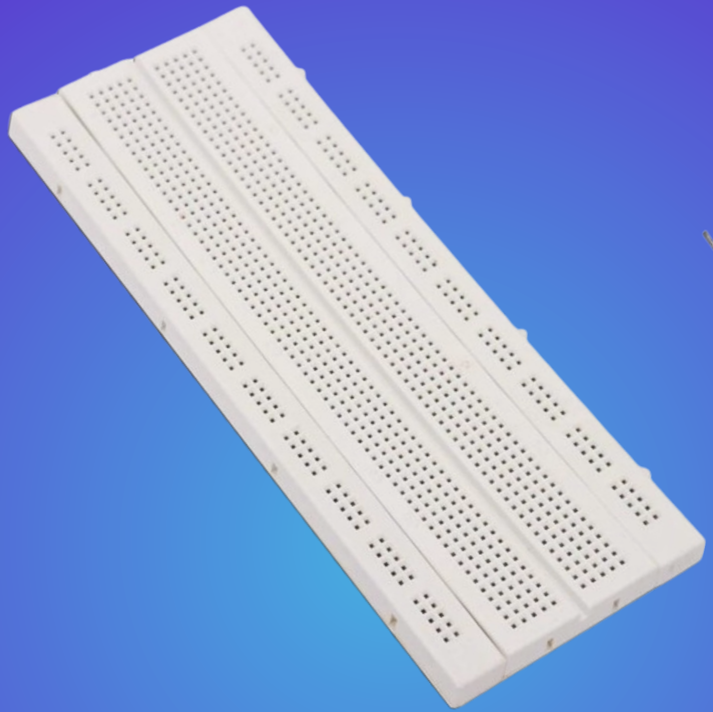
High sensitivity to ammonia gas, sulfide, benzene series steam, also can monitor smoke and other toxic gases well.



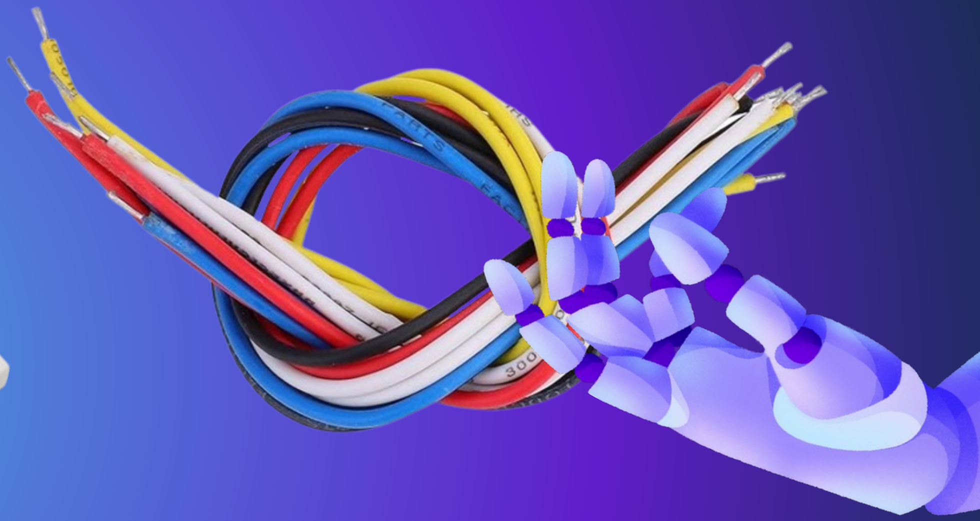
# NodeMCU – ESP8266

System on a chip (SOC) Wi-Fi microchip for Internet of Things (IoT) applications.

Jumper Wires



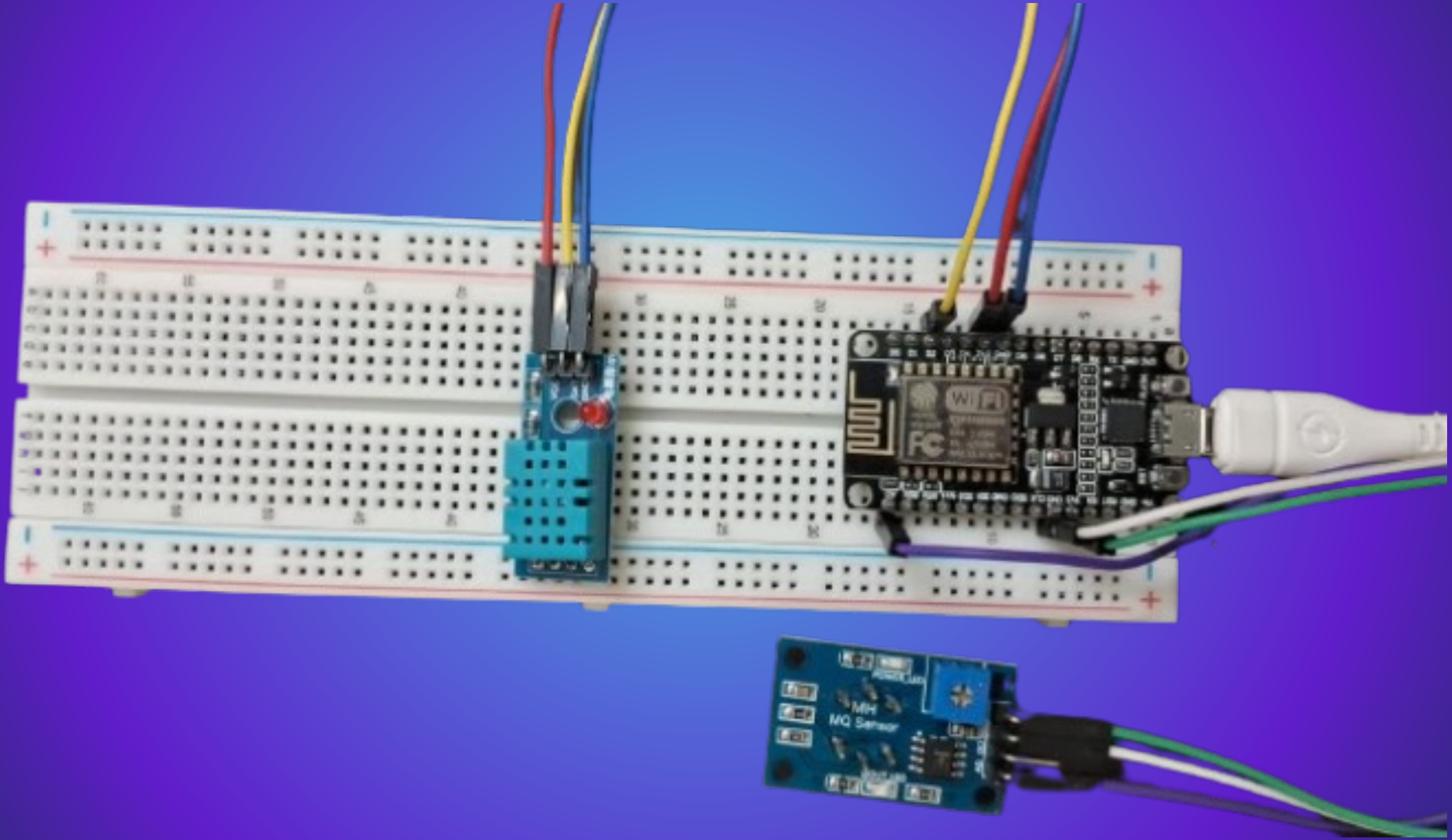
Breadboard



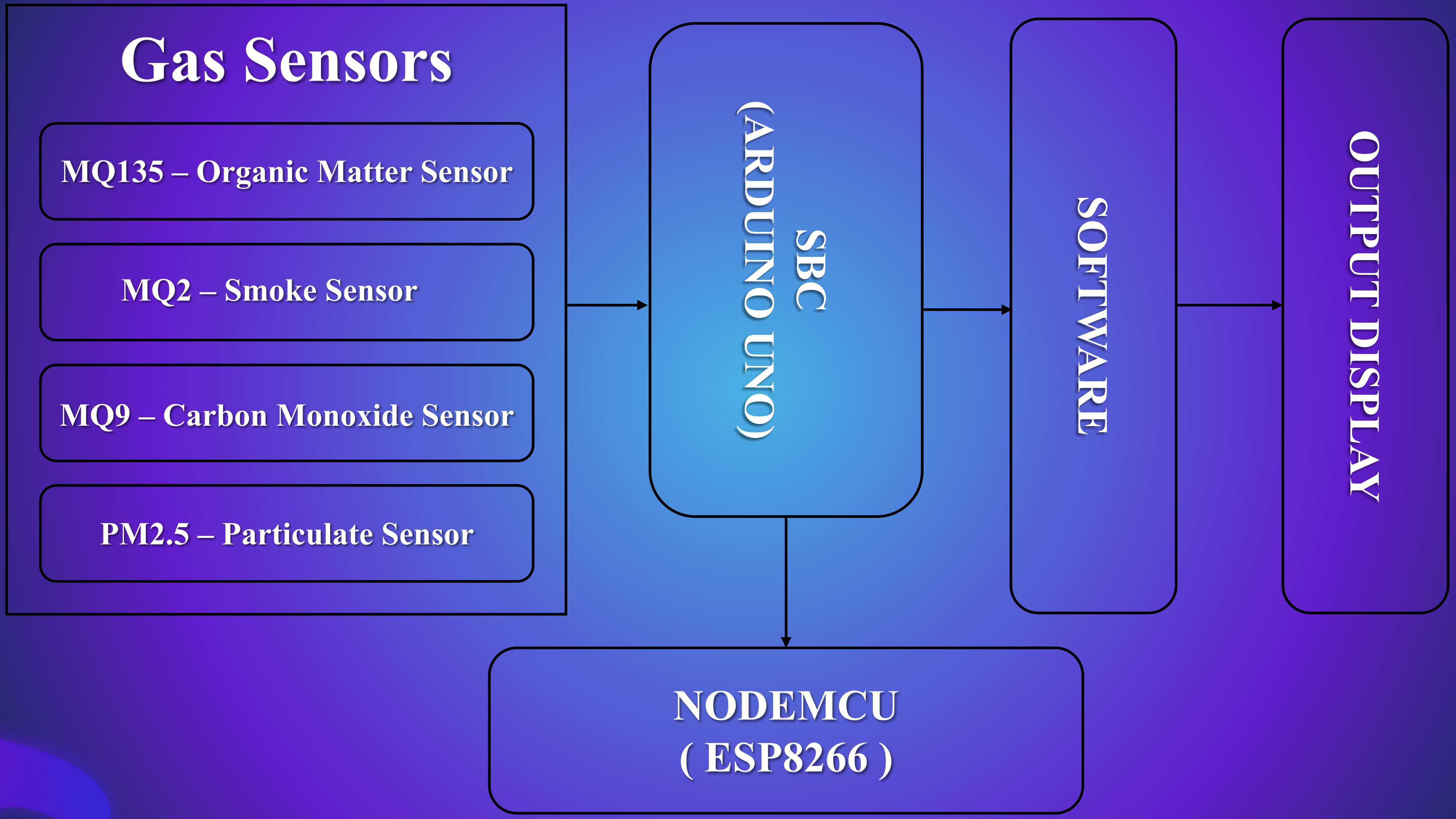
Connecting Wires

# DESIGN PROCESS

# CIRCUIT / BLOCK DIAGRAM



# OPERATION / FLOW DIAGRAM



# FUNCTIONALITY AND FEATURES

- Real-time Monitoring: The system continuously collects data from the sensors ,providing up-to-date information about temperature, humidity, and air quality parameters.
- Data Logging: It can log the collected data over time, allowing users to analyze historical trends and patterns in air quality.
- Cloud Integration: Integration with cloud services enables users to access monitoring data from anywhere via web or mobile applications.
- Dashboard Visualization: Arduino IoT Cloud provides customizable dashboards for visualizing sensor data in a user-friendly interface. Users can create graphs, charts, and widgets to display the information according to their preferences.

# APPLICATIONS

- Used to check the quality of air present indoor and outdoor. It can be used as safety monitoring system in the indoor spaces.
- It can be effectively used in hospitals and chemical research centers to prevent leakage of toxins into the air and help in immediate evacuation if necessary.
- It can be used to check the emissions of cars, automobiles and other vehicles and control pollution in the traffic accordingly.
- Information on AQ can be used to study the effects of climate change on air quality.

# FUTURE IMPROVEMENTS

- Future smart home systems might employ this air quality monitoring technology to continually monitor interior air quality by detecting the quantity of various contaminants and, as a result, triggering mechanical ventilation systems before they reach unsafe levels.
- Develop a model which predicts the quality of air based on historical data, weather patterns ,etc.
- Use of more accurate and sensitive sensor which are capable of detecting more number of pollutants in real time.

# THANK YOU